MIL-C-63112B (AR)
21 May 1986
SUPERSEDING
MIL-C-63112A (AR)
22 May 1981

MILITARY SPECIFICATION

CARTRIDGE 22MM, SUB-CALIBER PRACTICE M744, M745, M746, M747 LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the manufacture of metal parts, loading, assembling and packing of one type of training round for the 60MM and 81MM Mortar System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-48077 - Primer, M104 Parts for and Loading, Assembling and Packing
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC: 1305

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of the specification to the extent specified herein:

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

9299079 - Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747
9322198 - Tray Assembly for Cartridge, Sub-Caliber, 22MM, Practice
9322199 - Bag, Barrier
9322201 - Box, Wirebound
9322202 - Packing and Marking For Box, Wood, Wirebound, For Cartridge, Sub-Caliber 22MM, Practice

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.1.1 Nondestructive verification of the presence of the steel fuze locking balls. (see 4.4.2.22) Either of two inspection methods are required to be used for this verification:

a. An increase in the attractive force of a permanent magnet probe held in close proximity to the cartridge case.

b. A distinct indication on the meter of a commercially available eddy current crack detector, the probe of which was located similarly.

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For either method the following is required:

(1) Calibration of the system using known standards

(2) Permanent marking of satisfactory cartridges that is checked 100% by automated equipment before packaging.

3.2 Ejection and propelling charges and components and assemblies

3.2.1 Ejection and propelling charge. The ejection and propelling charges shall comply with all requirements specified on drawing (dwg.) 9299080 or 9331155 as applicable and associated drawings and with all requirements specified in applicable specifications and standards.

3.2.1.1 Propelling and ejection charge functioning

a. The propelling charge and the ejection charge for charge one (ref. part no. 9299080-1) shall produce a velocity of no more then 158 feet per second (FPS) nor less then 130 FPS to launch the 81MM sabot assembly at a quandrant elevation (QE) of 800 mils.

b. The propelling charge and the ejection charge for charge two (ref. part no 9299080-2) shall produce a velocity of no more than 174 FPS nor less than 144 FPS to launch the 81MM sabot assembly at a QE of 800 mils.

c. The propelling charge and the ejection charge for charge three (ref. part no. 9299080-3) shall produce a velocity of no more than 215 FPS nor less than 177 FPS to launch the 81MM sabot assembly at a QE of 800 mils.

d. The propelling charge and the ejection charge for charge four (ref. part no. 9299080-4) shall produce a velocity of no more than 245 FPS nor less than 202 FPS to launch the 81MM sabot assembly at a QE of 800 mils

e. The sabot assembly shall impact within an area of five by fifteen meters placed two meters from the mortar tube after the cartridge and sabot assembly have been fired.

3.2.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9299079 and associated drawings and with all requirements specified in applicable specifications and standards.

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3.3 Proving ground test. The cartridges shall comply with the following requirements when tested in accordance with 4.5.2 and subjected to ambient (70°F) temperature condition.

3.3.1 Misfires. The cartridges shall not fail to fire from the weapon.

3.3.2 Duds. The cartridges shall function on impact. The cartridges upon initiation on impact shall issue an audible report and a visible smoke spotting cloud.

3.3.3 Short rounds. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range (calculated without the short round).

3.3.4 Premature burst. There shall be no premature burst in the gun bore or in flight.

3.3.5 Metal parts separation. The metal parts of the 22mm cartridge shall not separate in flight. (See para 4.3.3.).

3.3.6 Transportation vibration. The cartridges shall be safe and operable after being subjected to the procedures as specified in MIL-STD-810, Method 514, procedure X and XI, curve AA.

3.4 Cartridge functional assembly. The subcaliber cartridges shall chamber (without being forced) in a fixture simulating the maximum metal condition of an M1 or M3 sabot.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract. (See para. 4.3.3.)

3.6 Workmanship. All parts shall be finished and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provision of 4.3.2. The first article sample shall consist of 1 box (dwg. 9322201) with 25 each cartridges (dwg. 9299079) and components in the quantities indicated in Table 1.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection. See MIL-A-48078 and as follows for the first article ballistic testing for each type:

a. One or more misfires occur.

b. One or more short rounds occur.

c. One or more premature bursts occur.

d. Two or more duds occur.

e. One or more 22 subcaliber metal parts separation occurs. Except for washer cup 9299109 and seal disc 9299104 which are normally expelled from the cartridge case at time of firing.

f. One or more sabots fail to exit the mortar tube.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747		1 OF 6		See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Head (Dwg. 9299110) Examination for defects	15		3.2	4.4.2.1
	Case, Cartridge Assembly (Dwg. 9299112) Examination for defects	15		3.2	4.4.2.2
	Case, Cartridge (Dwg. 9299105) Examination for defects	15		3.2	4.4.2.3
	Cup, Washer (Dwg. 9299109) Examination for defects	15		3.2	4.4.2.4
	Slide Percussion (Dwg. 9299102) Examination for defects	15		3.2	4.4.2.6
	Case Assembly (Dwg. 9299114) Examination for defects	15		3.2	4.4.2.7

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747	2	6	See Below
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case, Shell (Dwg. 9311005) Examination for defects	15		3.2	4.4.2.8
	Disc, Closing (Dwg. 9311004) Examination for defects	15		3.2	4.4.2.9
	Wing Assembly (Dwg. 9311002) Examination for defects	15		3.2	4.4.2.10
	Wing (Before forming) (Dwg. 9311003) Wing Embrittlement	15		3.2	4.5.4
	Disc - Non Metallic (Dwg. 9391009) Examination for defects	15		3.2	4.4.2.11
	Wing (After forming) (Dwg. 9311003) Examination for defects	15		3.2	4.4.2.12

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 3 OF 6		DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Case Ignition (Dwg. 9299108) Examination for defects	15		3.2	4.4.2.13
	Basin (Type 1, 2 & 3) (Dwg. 9299095) Examination for defects	15		3.2	4.4.2.14
	Basin, Type 4 (if applicable) (Dwg. 9299100) Examination for defects	15		3.2	4.4.2.15
	Nozzle, Jet (Dwg. 9299089) Examination for defects	15		3.2	4.4.2.16
	Disc (Dwg. 9299090) Examination for defects	15		3.2	4.4.2.17
	Foil, Damming (Dwg. 9299093) Examination for defects	15		3.2	4.4.2.18

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			SHEET 4 OF 6	DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Jet Nozzle Assembly (Dwg. 9299088)				
	Examination for defects	15		3.2	4.4.2.19
	Jet Housing Assembly (Type 1, 2 & 3) (Dwg. 9299080)				
	Examination for defects	15		3.2	4.4.2.20
	Charge weight assessment (for each individual type)	30		3.2	See Para. 3.2.1.1
	Cap, Protective (Dwg. 9369942)				
	Examination for defects	15		3.2	4.4.2.21
	Cartridge, 22MM Sub-Caliber Practice M774, M745, M746, M747 (Dwg. 9299079)				
	Examination for defects	25 ea		3.2	4.4.2.22
	Proving ground test	20 ea type		3.2	4.4.2.34 (a) (b)
	Transportation vibration test	25 ea type		3.3.5	4.5.2.2
	Body, Jet (Dwg. 9299084)				
	Examination for defects	15		3.2	4.4.2.24

NOTES: (a) The Proving Ground Test shall be performed on the same units that were subjected to the transportation vibration test.

(b) The excess units subjected to TV Test shall be placed in stock or utilized for conditioning/spotting rounds.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 5 OF 6		DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Charge, Ejection (Dwg. 9299098) Examination for defects	4 grams		3.2	4.4.2.25
	Anticreep Device (Dwg. 9369943) Examination for defects	15		3.2	4.4.2.26
	Cap, Percussion (Dwg. 9299103) Examination for defects	15		3.2	4.4.2.27
	Disc, Seal (Dwg. 9299104) Examination for defects	15		3.2	4.4.2.28
	Disc (Dwg. 9299107) Examination for defects	15		3.2	4.4.2.29
	Smoke Charge Assembly (Dwg. 9299111) Examination for defects	15		3.2	4.4.2.30

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 6 OF 6		DRAWING NUMBER
	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Charge, Smoke (Dwg. 9299113)				
	Examination for defects	14 grams		3.2	4.4.2.31
	Jet Housing Assembly (Type 4) (Dwg. 9331155)				
	Examination for defects	15		3.2	4.4.2.32
	Charge weight assessment	30		3.2	See Para. 3.2.1.1

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Cartridges shall contain:

a. Metal parts from lots with the same interfix lot number from one manufacturer.

b. Primers from lots with the same interfix lot number from one manufacturer.

c. Smoke charge, propelling charge, and ejection charge from lots with the same interfix lot numbers from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Head				9299110
					NEXT HIGHER ASSEMBLY
					9299079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Pitch diameter of rear threads		0.40%	3.2	Gage
103	Major diameter of rear threads		0.40%	3.2	Gage
104	Runout of rear threads with outside diameter and shoulder face forward of threads		0.40%	3.2	Gage
105	Diameter of groove		0.40%	3.2	Gage
106	Depth to shoulder forward of threads		0.40%	3.2	Gage
107	Profile of firing pin		0.40%	3.2	Gage
108	Length of protrusion of firing pin beyond aft face		0.40%	3.2	Gage
109	Depth of cavity		0.40%	3.2	Gage
110	Diameter of cavity, minimum		0.40%	3.2	Gage
111	Diameter of cavity chamfer angle		0.40%	3.2	Gage
112	Radius of firing pin point		0.40%	3.2	Gage
Minor					
201	Protective finish damaged or inadequate		0.65%	3.2	Visual
202	Width of groove		0.65%	3.2	Gage
203	Chamfer angle of cavity		0.65%	3.2	Gage
204	Diameter of firing pin base		0.65%	3.2	Gage
205	Overall length of head		0.65%	3.2	Gage
206	Taper of ogive		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Case, Cartridge Assembly				9299112
					NEXT HIGHER ASSEMBLY 9299079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max.		0.40%	3.2	Gage
102	Crimped inner diameter, max.		0.40%	3.2	Gage
103	Crimp inadequate or cracked		0.40%	3.2	Visual
104	Component parts missing or improperly assembled		0.40%	3.2	Visual
105	Diameter of balls		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Case, Cartridge				9299105
					NEXT HIGHER ASSEMBLY 9299112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Large cavity inside diameter		100%	3.2	Gage
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Thickness of forward crimping surface		0.40%	3.2	Gage
103	Pitch diameter thread		0.40%	3.2	Gage
104	Minor diameter thread		0.40%	3.2	Gage
105	Length, overall		0.40%	3.2	Gage
106	Runout of rear thread outside diameter and aft face		0.40%	3.2	Gage
107	Length of unthreaded inside chamber		0.40%	3.2	Gage
108	True position of large cavity inside diameter with outside diameter		0.40%	3.2	Gage
Minor					
201	Radius at aft end of crimping surface improper or missing		0.65%	3.2	Gage
202	Length from forward face to aft end of radius at outside surface		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Cup, Washer				9299109
					NEXT HIGHER ASSEMBLY 9299112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Height		0.40%	3.2	Gage
103	Wall thickness at wall base		0.40%	3.2	Gage
Minor					
201	Thickness of side wall		0.65%	3.2	Gage
202	Inside radius missing or improper		0.65%	3.2	Visual
203	Outside radius missing or improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299101
4.4.2.5	Percussion Slide Assembly				NEXT HIGHER ASSEMBLY 9299112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Primer missing or deformed		0.40%	3.2	Visual
102	Components improperly assembled		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Slide, Percussion				9299102
					NEXT HIGHER ASSEMBLY 9299101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Outside diameter		100%	3.2	Gage
B	Angle of cutout groove		100%	3.2	Gage
Major					
101	Location of outside ball radius groove		0.40%	3.2	Gage
102	Inside diameter of thru hole		0.40%	3.2	Gage
103	Location of ball-groove apex		0.40%	3.2	Gage
104	Aft outside diameter		0.40%	3.2	Gage
105	Length from aft face to step		0.40%	3.2	Gage
106	True position of aft outside diameter with outside diameter and step		0.40%	3.2	Gage
107	Primer seating chamfer improper		0.40%	3.2	Gage
Minor					
201	Diameter of large cavity		0.65%	3.2	Gage
202	Depth of large diameter cavity		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Case Assembly				9299114
					NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max		0.40%	3.2	Gage
102	Component parts missing or improperly assembled		0.40%	3.2	Visual
103	Length from end of shell case to wing		0.40%	3.2	Gage
104	Runout of aft diameter with forward outside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.8	Case, Shell				9311005
					NEXT HIGHER ASSEMBLY
					9299114
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thread pitch diameter		0.40%	3.2	Gage
102	Thread minor diameter		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
104	Inside diameter (wing end)		0.40%	3.2	Gage
105	Runout of rear inside diameter to outside diameter		0.40%	3.2	Gage
106	Runout of thread to outside diameter and forward face		0.40%	3.2	Gage
107	Length, overall		0.40%	3.2	Gage
108	Length from aft face to aft end of wing interface groove		0.40%	3.2	Gage
109	Length from aft face to forward end of rear cavity		0.40%	3.2	Gage
110	Depth of wing interface groove		0.40%	3.2	Gage
Minor					
201	Protective finish missing or inadequate		0.65%	3.2	Visual
202	Central inside diameter		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	Case, Shell				9311005
					NEXT HIGHER ASSEMBLY
					9299114
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	Runout of central inside diameter with outside diameter		0.65%	3.2	Gage
204	Forward cavity diameter aft of the thread		0.65%	3.2	Gage
205	Length from forward face to aft end of forward cavity		0.65%	3.2	Gage
206	Depth of forward cavity		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Disc, Closing				9311004
					NEXT HIGHER ASSEMBLY 9299114
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Protective finish missing or inadequate		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Wing Assembly				9311002
					NEXT HIGHER ASSEMBLY 9299114
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Rear inside diameter		0.40%	3.2	Gage
103	Distance from front face to wing		0.40%	3.2	Gage
104	Forward outside diameter		0.40%	3.2	Gage
105	Runout of rear face to forward outside diameter		0.40%	3.2	Gage
106	Perpendicularity between aft face and forward outside diameter		0.40%	3.2	Gage
107	Inside diameter of cavity		0.40%	3.2	Gage
108	Distance from ends of parallel fin blades		0.40%	3.2	Gage
109	Fin blades, damaged, distorted, or improperly molded to body		0.40%	3.2	Visual
Minor					
201	Depth of cavity		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Disc - Nonmetallic				9391009
					NEXT HIGHER ASSEMBLY 9299080/9331155
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
Minor					
201	Poor workanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9311003
4.4.2.12	Wing (After forming)				NEXT HIGHER ASSEMBLY 9311002
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wings not normal to each other		0.40%	3.2	Gage
102	Length from kink to end of wing		0.40%	3.2	Gage
103	Angle of kink		0.40%	3.2	Gage
104	Length of kinked wing		0.40%	3.2	Gage
105	Length of unkinked wing		0.40%	3.2	Gage
106	Width of wing		0.40%	3.2	Gage
107	Thickness		0.40%	3.2	Gage
108	Angular cutoffs improperly located or incorrect (4 places)		0.40%	3.2	Gage
109	Radii missing or improper (4 places)		0.40%	3.2	Visual
110	Presence of holes		0.40%	3.2	Visual
Minor					
201	Kink radius missing or improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9299108
4.4.2.13	Case, Ignition				NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter, internal thread		0.40%	3.2	Gage
102	Minor diameter, internal thread		0.40%	3.2	Gage
103	Pitch diameter, external thread		0.40%	3.2	Gage
104	Major diameter, external thread		0.40%	3.2	Gage
105	Diameter (2 holes)		0.40%	3.2	Gage
106	Location of holes		0.40%	3.2	Gage
107	Inside diameter of large cavity		0.40%	3.2	Gage
108	Outside diameter, max.		0.40%	3.2	Gage
109	Runout of outside thread with outside diameter and shoulder forward of aft thread		0.40%	3.2	Gage
110	Runout of inside thread with outside diameter and shoulder forward of aft thread		0.40%	3.2	Gage
111	Total length		0.40%	3.2	Gage
112	Diameter of inside cavity groove		0.40%	3.2	Gage
113	Diameter of aft funnel cavity		0.40%	3.2	Gage
114	Length from aft face to shoulder		0.40%	3.2	Gage
115	Width of inside cavity groove		0.40%	3.2	Gage
116	Length from forward face to aft end of inside cavity groove		0.40%	3.2	Gage
117	Inside diameter of small cavity		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.13	Case, Ignition				9299108
					NEXT HIGHER ASSEMBLY
					9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
118	Length from forward face to aft end of small cavity		0.40%	3.2	Gage
119	Length from forward face to aft end of large cavity		0.40%	3.2	Gage
Minor					
201	Runout of forward cavity diameter with outside diameter		0.65%	3.2	Gage
202	Runout of aft tunnel cavity with outside diameter		0.65%	3.2	Gage
203	Diameter of case forward of aft thread		0.65%	3.2	Gage
204	Inside cavity groove radii missing or improper		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Basin (Type 1, 2, & 3)				9299095
					NEXT HIGHER ASSEMBLY
					9299080
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
Minor					
201	Radius missing or incomplete		0.65%	3.2	Visual
202	Wall thickness		0.65%	3.2	Gage
203	Height		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Basin (Type 4)				9299100
					NEXT HIGHER ASSEMBLY 9331155
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of raised hat		0.40%	3.2	Gage
Minor					
201	Presence of radii		0.65%	3.2	Visual
202	Wall thickness		0.65%	3.2	Gage
203	Height		0.65%	3.2	Gage
204	Outside diameter		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299089
4.4.2.16	Nozzle, Jet				NEXT HIGHER ASSEMBLY 9299088
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter, thread		0.40%	3.2	Gage
102	Major diameter, thread		0.40%	3.2	Gage
103	True position of holes (8 holes)		0.40%	3.2	Gage
104	Runout of inner diameter with forward face and thread		0.40%	3.2	Gage
105	Inner diameter		0.40%	3.2	Gage
106	Diameter of holes		0.40%	3.2	Gage
107	Inside diameter of percussion cap cavity		0.40%	3.2	Gage
108	Aft outside diameter		0.40%	3.2	Gage
109	Depth of percussion cap cavity		0.40%	3.2	Gage
110	Length from forward face to seat		0.40%	3.2	Gage
111	Overall length		0.40%	3.2	Gage
Minor					
201	True position of percussion cap cavity with inside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Disc				9299090
					NEXT HIGHER ASSEMBLY
					9299088
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of boss		0.40%	3.2	Gage
102	Height of boss		0.40%	3.2	Gage
Minor					
201	Thickness		0.65%	3.2	Gage
202	Diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Foil, Damming	1 OF 1	9299093
			NEXT HIGHER ASSEMBLY: 9299088

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Adhesive coating missing or incomplete		0.40%	3.2	Visual
102	Inner diameter		0.40%	3.2	Gage
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Thickness of foil		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Jet Nozzle Assembly				9299088
					NEXT HIGHER ASSEMBLY
					9299080, 9331155
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Crimp improper or missing		0.40%	3.2	Visual
102	Disc insecure or missing		0.40%	3.2	Visual
103	Cap missing, deformed or loose		0.40%	3.2	Visual
104	Foil missing, deformed or loose		0.40%	3.2	Visual
105	Lacquer not complete for 360°		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Jet Housing Assembly (Type 1, 2, & 3)				9299080
					NEXT HIGHER ASSEMBLY 9299079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propelling charge not present		100%	3.2	Visual
102	Ejection charge not present		100%	3.2	Visual
103	Components improperly seated, loose, missing or deformed		100	3.2	Visual/Manual
104	Lacquer missing or improperly applied		0.40%	3.2	Visual
105	Torque improper		0.40%	3.2	Visual
106	Protective cap improperly assembled to jet housing		0.40%	3.2	Visual
107	Notches-not proper number for given charge		100%	3.2	Visual
108	Basin not flush or below flush with jet body			3.2	Visual
Minor					
201	Excessive adhesive (jet screw threads)		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.21	Cap, Protective				9369942
					NEXT HIGHER ASSEMBLY
					9299080/9331155
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of cavity		0.40%	3.2	Gage
102	Depth of cavity		0.40%	3.2	Gage
Minor					
201	Presence of finger grips		0.65%	3.6	Visual
202	Presence of flash inside cavity		0.65%	3.6	Visual
203	Height of cap		0.65%	3.2	Gage
204	Rear outside diameter (hub)		0.65%	3.2	Gage
205	Forward outside diameter		0.65%	3.2	Gage
206	Cavity radius		0.65%	3.2	Gage
207	Cavity radius missing or improper		0.65%	3.2	Visual
208	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.22	Cartridge 22MM, Sub-Caliber Practice M744, M745, M746, M747		1 OF 1	9299079
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety balls missing		100%	3.1.1	Gage
Special					
A	Anti creep device missing		100%	3.2	Visual
Major					
101	Cartridge does not go into or fully seat in a fixture simulating a max material sabot chamber		100%	3.2	Gage
102	Forward outside diameter, max.		0.40%	3.2	Chamber gage
103	Jet housing assembly torque		0.40%	3.2	Gage
104	Head torque		0.40%	3.2	Gage
105	Overall length, max.		0.40%	3.2	Gage
106	True position of jet housing assembly barrel outside diameter to case cartridge assembly outside diameter		0.40%	3.2	Gage
107	Head not painted or improperly painted		0.40%	3.2	Visual
108	Marking missing or incomplete		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Tray Assembly for Cartridge Sub-Caliber, 22MM Practice				9322198
					NEXT HIGHER ASSEMBLY 9322202
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cover and body do not mate		0.40%	3.2	Visual
102	Assembly damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Body, Jet	1 OF 2	9299084
			NEXT HIGHER ASSEMBLY 9299080/9331155

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Major diameter of forward thread		0.40%	3.2	Gage
103	Pitch diameter of forward thread		0.40%	3.2	Gage
104	Minor diameter of aft thread		0.40%	3.2	Gage
105	Pitch diameter of aft thread		0.40%	3.2	Gage
106	Runout of aft thread with diameter forward of shank and aft face		0.40%	3.2	Gage
107	Runout of forward thread with diameter forward of shank and aft thread relief surface		0.40%	3.2	Gage
108	Outside diameter of shank		0.40%	3.2	Gage
109	Outside diameter forward of shank		0.40%	3.2	Gage
110	Outside diameter aft of shank		0.40%	3.2	Gage
111	Length from forward face to first step		0.40%	3.2	Gage
112	Length from forward face to large diameter shank step		0.40%	3.2	Gage
113	Length from aft face to large diameter shank step		0.40%	3.2	Gage
114	Diameter of tunnel		0.40%	3.2	Gage
115	Length of forward cavity		0.40%	3.2	Gage
116	Length of aft cavity		0.40%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.24	Body, Jet				9299084
					NEXT HIGHER ASSEMBLY
					9299080/9331155
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Length of forward thread relief		0.65%	3.2	Gage
202	Presence of aft face angled notch		0.65%	3.2	Gage
203	Diameter of forward cavity		0.65%	3.2	Gage
204	Forward or aft thread chamfers missing		0.65%	3.2	Gage
205	Diameter of aft cavity		0.65%	3.2	Gage
206	Diameter of aft cavity groove		0.65%	3.2	Gage
207	Length of aft cavity groove		0.65%	3.2	Gage
208	Length from aft face to forward edge of aft cavity groove		0.65%	3.6	Gage
209	Forward thread relief diameter		0.65%	3.2	Gage
210	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Charge, Ejection	1 OF 1	9299098
			NEXT HIGHER ASSEMBLY: 9331155/9299080

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Composition improper	2 grams		3.2	Chemical Analysis
102	Granulation size improper	2 grams		3.2	Gage
Minor					
201	Evidence of poor workmanship	2 grams		3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.26	Anti Creep Device				9369943
					NEXT HIGHER ASSEMBLY 9299079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect length		0.40%	3.2	Gage
102	Walls cracked or crinkled		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299103
4.4.2.27	Cap, Percussion				NEXT HIGHER ASSEMBLY 9299101
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter at flange		0.40%	3.2	Gage
102	Outside diameter of primer cup forward of flange		0.40%	3.2	Gage
103	Overall length		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9299104
4.4.2.28	Disc, Seal				NEXT HIGHER ASSEMBLY 9299112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of disc		0.40%	3.2	Gage
Minor					
201	Thickness of disc		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.29	Disc				9299107
					NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of disc		0.40%	3.2	Gage
Minor					
201	Diameter of disc		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.30	Smoke Charge Assembly		9299111
			NEXT HIGHER ASSEMBLY 9299112

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Ignition case improperly torqued to case assembly		0.40%	3.2	Gage
103	Improper assembly of components		0.40%	3.2	Visual
104	Smoke charge missing		0.40%	3.2	Radiographic
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.31	Charge, Smoke				9299113
					NEXT HIGHER ASSEMBLY 9299111
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Mixture improper	20 grams		3.2	Chemical Analysis
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9331155
4.4.2.32	Jet Housing Assembly (Type 4)				NEXT HIGHER ASSEMBLY 9399079
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propelling charge not present		100%	3.2	Visual
102	Ejection charge not present		100%	3.2	Visual
103	Components improperly seated, loose, missing or deformed		100%	3.2	Visual/Manual
104	Lacquer missing or improperly applied		0.40%	3.2	Visual
105	Torque improper		0.40%	3.2	Visual
106	Protective cap improperly sealed		0.40%	3.2	Visual
107	Notches - proper number		100%	3.2	Visual
108	Basin flush or below flush with jet body		0.40%	3.2	Visual
Minor					
201	Excessive adhesive (jet screw threads)		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-63112B (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.33	Packing and Marking for Box, Wood, Wirebound, for Cartridge, Subcaliber 22MM, Practice		1 OF 1	9322202
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged		0.40%	3.2	Visual
Minor					
201	Marking missing, illegible, or incomplete		0.65%	3.2	Visual
202	Barrier bag torn or improperly sealed		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.34	Proving Ground Test Cartridge 22MM, Sub-Caliber Practice, M744, M745, M747, M747	1 OF 1	9299079
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST (c) (d)	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special A	Short round	20	(a)	3.3	4.5.2
Major					
101	Misfire	20	(b)	3.3	4.5.2
102	Cartridge fails to function audibly	20	(b)	3.3	4.5.2
103	Smoke charge failure	20	(b)	3.3	4.5.2
104	Premature burst	20	(b)	3.3	4.5.2
105	Duds	20	(b)	3.3	4.5.2
106	Velocity	20	(b)	3.3	4.5.2
107	Metal parts separation	20	(a) (e)	3.3	4.5.2
Minor					
201	Sabot does not impact in required landing area	20	(b)	3.2.1	4.5.2

NOTES: (a) One or more defects shall be cause for lot rejection. Wind speed shall be less than 10 MPH at time of firing. (b) Two or more defects shall be cause for lot rejection. (c) A total of twenty units shall be fired for this test. Fired units shall be examined for all the above defects for each shot. (d) Prior to the ballistic test a soil penetration test shall be performed as per paragraph 4.5.2.5. (e) See para. 3.3.5.

AMSMC Form 1570a, 1 Apr 85 Replaces DRDAR-QA Form 160a, 1 Jun 83, which may not be used.

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4.4.3 Testing. See 4.5.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5 METHODS OF INSPECTION

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspections and test data to determine that all components (parts, sub-assemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table 1.

4.5.2 Proving ground test

4.5.2.1 Functioning test. The test shall be performed at a Government Proving Ground in accordance with the applicable USATECOM Acceptance Test Procedure for standard 60MM or 81MM mortar testing. A standard soil penetrometer test (see para. 4.5.2.5) shall be performed on the impact area just prior to ballistic testing. Use six sabots in succession. Sabots will not require conditioning. The cartridges shall be conditioned a minimum of six (6) hours at 70°F and test fired within two (2) minutes after removal from the conditioning chamber. The individual velocities, deflections, ranges, sound and smoke functioning shall be recorded on the firing record. Temperature of the day, wind and weather conditions shall also be recorded. Production lot testing shall utilize twenty (20) rounds at each charge level (1 to 4) with a firing elevation of 800 mils.

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4.5.2.2 Transportation-vibration. The cartridges in the packaged condition shall be subjected to the required vibration test, (see para. 3.3.5) conditioned at range ambient temperature and test fired per the procedure of 4.5.2.1. (First article only.)

4.5.2.3 Cleaning cannon M224, M29A1 or M29. Before firing apply rifle bore cleaner, solvent cleaning compound to wiping cloths gripped in proper cleaning staff. Insert it into bore and thoroughly scrub all surfaces with a push-pull action the entire length of barrel. Make certain to clean firing pin using clean wiping cloths on cleaning rod and dry the bore thoroughly. Examine firing pin for cleanliness and that the pin is free of residue during firing. After every tenth firing, the bore should be cleaned with rifle bore cleaner and wiped dry with clean wiping cloths.

4.5.2.4 Cleaning M1 sabot. Before firing, the sabot is to be cleaned using a cleaning wick positioned in the slot of the end staff and saturated with rifle bore cleaner. Then dry thoroughly with a clean dry wick. During firing, the M1 Sabot should be cleaned after each round is fired with a water detergent solution to remove any dirt or dust, then wiped dry including the bore.

4.5.2.5 Test site impact area soil condition (soil penetrometer test). Just before each ballistic firing, the impact area shall be tested for stiffness at the ranges of 200, 250, 300 and 400 meters using a 15 pound slide hammer penetrometer with a half inch cone. In dirt and clay the number of blows per foot shall be between 9 and 15 inclusive. The penetration test shall be performed at the 12, 3, 6 and 9 o'clock position as seen from the weapon. These tests shall be spaced approximately six feet apart.

4.5.3 Check test for deterioration of primers. If the total time between original acceptance of any primer lot and the assembly of that lot into the cartridges exceeds two years, at any time since previous tests, the primer lot shall be subjected to the functioning and output tests specified in MIL-P-48077 immediately before the primer lot is assembled into the cartridges. The primers must meet the functioning and output criteria to the original manufacturer (see dwg 9299103). This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the cartridges. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or the contractor who assembled the primers into the cartridges. The test shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct

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result of carelessness in handling, storage, etc., permitted while the primer was under the jurisdiction of either contractor.

4.5.4 Wing embrittlement. The wing shall be bent until it no longer returns to its original position. When the spring no longer returns to its original profile, it shall be considered distorted permanently. Any wing that breaks prior to this point shall be classed defective.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be as specified on dwg. 9322198, 9322199, and 9322201.

5.2 Packing and marking. Packing and marking shall be as specified on dwg. 9322202.

6. NOTES

6.1 Intended use. The cartridges covered by this specification are intended for use on the 60MM and 81MM Mortar Training Device.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAT-I, Dover NJ 07801-5001.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes from previous issue. Asterists (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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